Signals, Systems and Transforms

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Chapter 4

Discrete-Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Chapter Intended Learning Outcomes :

(i) Understanding the characteristics and properties of DTFT and DFT

(ii) Ability to perform discrete-time signal conversion between the time and frequency domains using DTFT and DFT and their inverse transforms

References

- J.G.Proakis and D.G.Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, Prentice Hall 2007.
- A.V.Oppenheim and R.W.Schafer, *Discrete-Time Signal Processing*, Prentice Hall, 1998.
- Hing Cheung So, *Lecture Notes on Digital Signal Processing,* Lectures at DEE of City University of Hong Kong, Feb. 2016.
- Mark Fowler, Digital Signal Processing, Lectures at Binghamton University, NY,2015

Part 1: Discrete-Time Fourier Transform (DTFT)

Definition

DTFT is a frequency analysis tool for aperiodic discrete-time signals

The DTFT of $x[n]$, $X(e^{j\omega})$, is given as

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \qquad (4.1)$$

Proof : A seen in Chapter 3, we construct the continuous-time sampled signal with a sampling interval of from :

$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT) \qquad (4.2)$$

Taking Fourier transform of $x_s(t)$ with using properties of $\delta(t)$:

$$X_s(j\Omega) = \int_{-\infty}^{\infty} x_s(t) e^{-j\Omega t} dt = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT) e^{-j\Omega t} dt$$

$$= \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega nT} \qquad (4.3)$$

Defining $\omega = \Omega T$ as the discrete-time frequency parameter and writing $X_s(j\Omega)$ as $X(e^{j\omega})$, (4.3) becomes

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

As in Fourier transform, $X(e^{j\omega})$ is also called spectrum and is a continuous function of the frequency parameter ω

To convert $X(e^{j\omega})$ to $x[n]$, we use inverse DTFT:

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega \tag{4.4}$$

Proof: Putting (4.1) into (4.4) :

$$\begin{aligned}
\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega &= \frac{1}{2\pi}\int_{-\pi}^{\pi}\left[\sum_{m=-\infty}^{\infty} x[m]e^{-j\omega m}\right]e^{j\omega n}d\omega \\
&= \frac{1}{2\pi}\sum_{m=-\infty}^{\infty} x[m]\int_{-\pi}^{\pi} e^{-j\omega m}e^{j\omega n}d\omega \\
&= \frac{1}{2\pi}\sum_{m=-\infty}^{\infty} x[m]\frac{2\sin((n-m)\pi)}{n-m} \\
&= x[n]
\end{aligned} \tag{4.5}$$

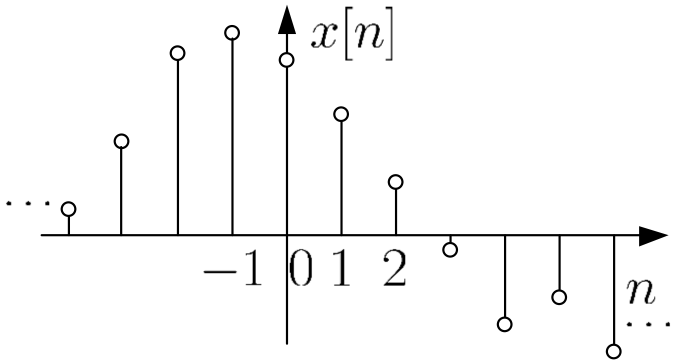
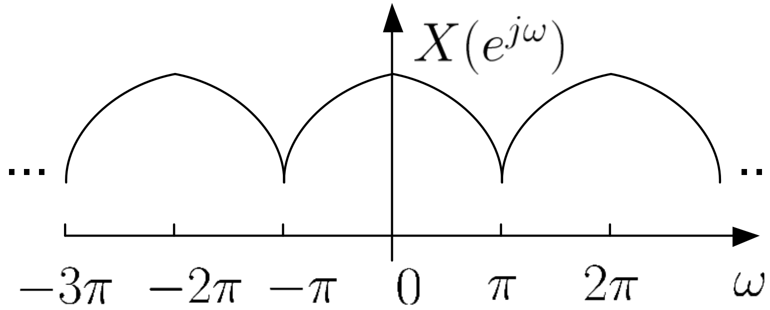
time domain	frequency domain
	
$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \Longrightarrow$	
	$\Longleftarrow x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$
discrete and aperiodic	continuous and periodic

Fig. 4.1 Illustration of DTFT

$X(e^{j\omega})$ is continuous and periodic with a period of 2π

$X(e^{j\omega})$ is generally complex, we can illustrate $X(e^{j\omega})$ using the magnitude and phase spectra, i.e., $|X(e^{j\omega})|$ and $\angle(X(e^{j\omega}))$:

$$|X(e^{j\omega})| = \sqrt{(\Re\{X(e^{j\omega})\})^2 + (\Im\{X(e^{j\omega})\})^2} \qquad (4.6)$$

and

$$\angle(X(e^{j\omega})) = \tan^{-1}\left(\frac{\Im\{X(e^{j\omega})\}}{\Re\{X(e^{j\omega})\}}\right) \qquad (4.7)$$

where both are continuous in frequency and periodic.

Convergence of DTFT

The DTFT of a sequence $x[n]$ converges if

$$\left|X(e^{j\omega})\right| = \left|\sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}\right| \leq \sum_{n=-\infty}^{\infty} |x[n]| \cdot \left|e^{-j\omega n}\right| = \sum_{n=-\infty}^{\infty} |x[n]| < \infty \quad (4.8)$$

Let $h[n]$ be the impulse response of a linear time-invariant (LTI) system, the following statements are equivalent:

S1. The system is stable so that $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$

S2. The DTFT of $h[n]$, i.e., $H(e^{j\omega})$, converges

Note that $H(e^{j\omega})$ is also known as system frequency response

<u>Example 4.1</u>

Determine the DTFT of $x[n] = u[n]$.

Using (4.1), the DTFT of $x[n]$ is computed as:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{\infty} e^{-j\omega n}$$

Since

$$\sum_{n=0}^{\infty} \left|e^{-j\omega n}\right| = \sum_{n=0}^{\infty} 1 = \infty$$

$X(e^{j\omega})$ does not exist.

Alternatively, employing the stability condition:

$$\sum_{n=-\infty}^{\infty} |u[n]| = \sum_{n=0}^{\infty} 1 = \infty$$

which also indicates that the DTFT does not converge

Example 4.2

Find the DTFT of $x[n] = u[n] - u[n-N]$. Plot the magnitude and phase spectra for $N = 10$.

Using (4.1), we have

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{N-1} e^{-j\omega n} = \frac{e^{-j\omega \cdot 0}\left(1 - e^{-j\omega N}\right)}{1 - e^{-j\omega}} = \frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}}$$

To plot the magnitude and phase spectra, we express $X(e^{j\omega})$:

$$X(e^{j\omega}) = \frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}} = \frac{e^{-j\omega N/2}}{e^{-j\omega/2}} \cdot \frac{e^{j\omega N/2} - e^{-j\omega N/2}}{e^{j\omega/2} - e^{-j\omega/2}} = e^{-j\omega(N-1)/2} \cdot \frac{\sin(\omega N/2)}{\sin(\omega/2)}$$

In doing so, $|X(e^{j\omega})|$ and $\angle(X(e^{j\omega}))$ can be written in closed-forms as:

$$|X(e^{j\omega})| = \left|\frac{\sin(\omega N/2)}{\sin(\omega/2)}\right|$$

and

$$\angle(X(e^{j\omega})) = -\frac{\omega(N-1)}{2} + \angle\left(\frac{\sin(\omega N/2)}{\sin(\omega/2)}\right)$$

Note that we generally employ (4.6) and (4.7) for magnitude and phase computation

In using MAT/OCT to plot $|X(e^{j\omega})|$ and $\angle(X(e^{j\omega}))$, we utilize

the command `sinc` so that there is no need to separately handle the "0/0" cases due to the sine functions

Recall the definition of sinc function: $\mathrm{sinc}(u) = \dfrac{\sin(\pi u)}{\pi u}$

As a result, we have:

$$\begin{aligned}\frac{\sin(\omega N/2)}{\sin(\omega/2)} &= \frac{\sin(\omega \cdot N\pi/(2\pi))}{\omega \cdot N\pi/(2\pi)} \cdot \frac{\omega \cdot N\pi}{2\pi} \cdot \frac{\omega\pi/(2\pi)}{\sin(\omega\pi/(2\pi))} \cdot \frac{2\pi}{\omega\pi} \\ &= N \cdot \frac{\operatorname{sinc}(\omega N/(2\pi))}{\operatorname{sinc}(\omega/(2\pi))}\end{aligned}$$

The key MATLAB code for $N = 10$ is

```
N=10;                   %N=10
w=0:0.01*pi:2*pi;       %successive frequency point
                        %separation is 0.01pi
dtft=N.*sinc(w.*N./2./pi)./(sinc(w./2./pi)).*exp(-
j.*w.*(N-1)./2);        %define DTFT function
subplot(2,1,1)
Mag=abs(dtft);          %compute magnitude
plot(w./pi,Mag);        %plot magnitude
subplot(2,1,2)
Pha=angle(dtft);        %compute phase
plot(w./pi,Pha);        %plot phase
```

201 uniformly-spaced points are taken to approximate the continuous functions $|X(e^{j\omega})|$ and $\angle(X(e^{j\omega}))$

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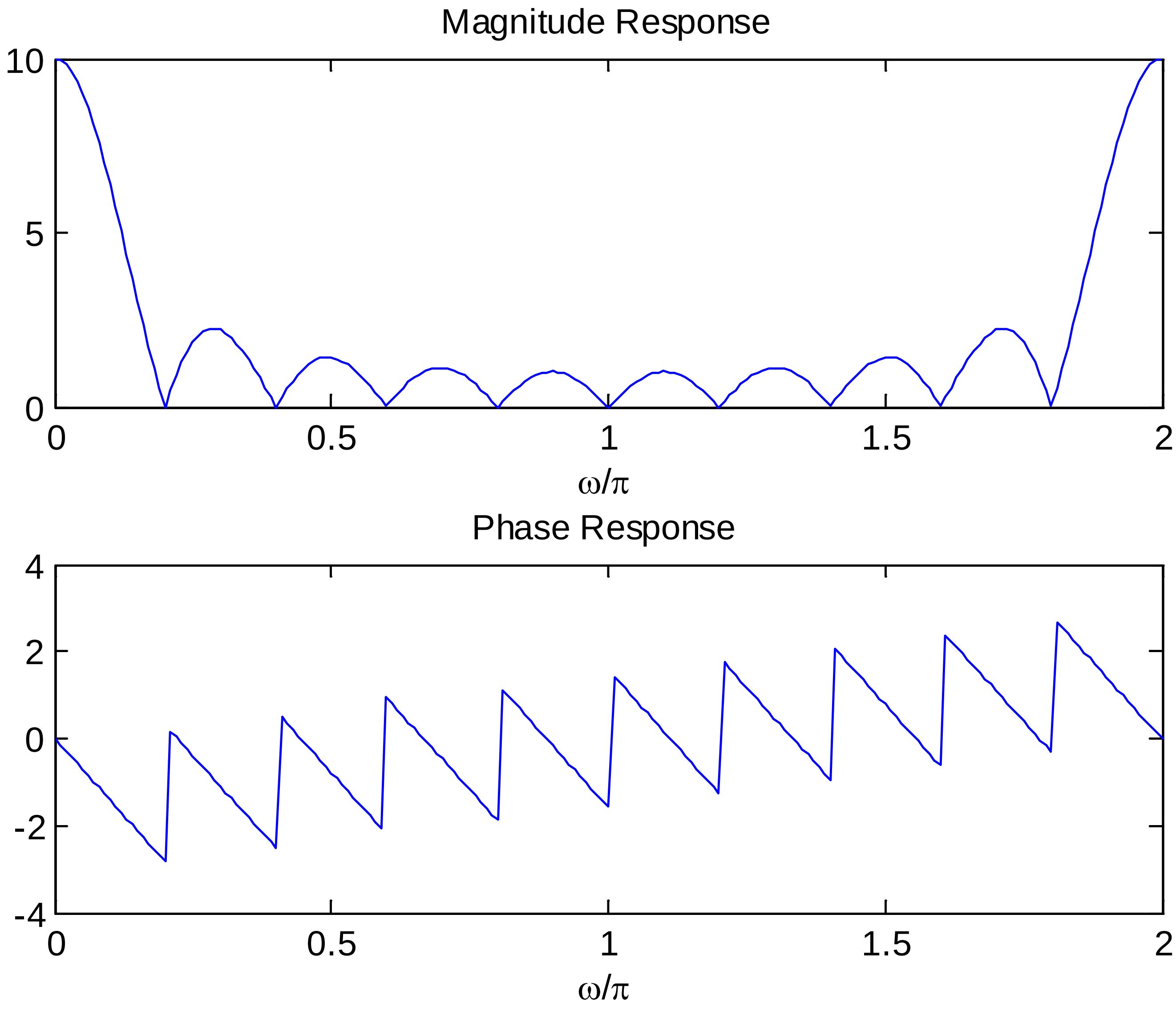

Fig. 4.2 DTFT plots using `abs` and `angle`

<u>Example 4.3</u>

Find the inverse DTFT of $X(e^{j\omega})$ which is a rectangular pulse within $\omega \in [-\pi, \pi]$:

$$X(e^{j\omega}) = \begin{cases} 1, & -w_0 < \omega < w_0 \\ 0, & \text{otherwise} \end{cases}$$

where $0 < w_0 < \pi$.

Using (4.4), we get:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-w_0}^{w_0} e^{j\omega n} d\omega = \frac{\sin(w_0 n)}{\pi n} = \frac{w_0}{\pi} \text{sinc}\left(\frac{w_0 n}{\pi}\right)$$

That is, $x[n]$ is an infinite-duration sequence whose values are drawn from a scaled sinc function.

Example 4.4

Determine the inverse DTFT of $X(e^{j\omega})$ which has the form of:

$$X(e^{j\omega}) = \frac{e^{-j2\omega}}{1 + 0.7e^{-j\omega}}$$

Employing the time shifting property, we get

$$x[n] = (-0.7)^{n-2}u[n-2]$$

Frequency analysis of LTI systems

Definition

For an LTI system with impulse response $h[n]$, the frequency response is the Fourier transform of $h[n]$:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}, \quad -\pi < \omega \leq \pi.$$

Eigenfunction Property

If the input is a complex exponential $x[n] = e^{j\omega_0 n}$, the output is:

$$y[n] = H(e^{j\omega_0})\, e^{j\omega_0 n}.$$

That is, $e^{j\omega_0 n}$ is an eigenfunction of the system, and $H(e^{j\omega_0})$ is the corresponding eigenvalue.

For Sinusoidal Inputs

If $x[n] = A\cos(\omega_0 n + \phi)$, then using Euler's formula and linearity:

$$y[n] = A|H(e^{j\omega_0})|\cos\left(\omega_0 n + \phi + \arg H(e^{j\omega_0})\right)$$

Remark: This relationship is also valid if we replace cosinus by sinus.

Magnitude response: $|H(e^{j\omega})|$ gain at frequency ω

Phase response: $\arg H(e^{j\omega})$ phase shift at frequency ω

Important Note

The frequency response completely characterizes the LTI system in the frequency domain. It tells us how the system modifies each frequency component of the input signal.

Properties of DTFT

Since DTFT is closely related to z-transform, its properties follow those of z-transform. Note that ROC is not involved because it should include unit circle in order for DTFT exists

1. Linearity

If $(x_1[n], X_1(e^{j\omega}))$ and $(x_2[n], X_2(e^{j\omega}))$ are two DTFT pairs, then:

$$ax_1[n] + bx_2[n] \leftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega}) \tag{4.8}$$

2. Time Shifting

A shift of n_0 in $x[n]$ causes a multiplication of $e^{-j\omega n_0}$ in $X(e^{j\omega})$:

$$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega}) \tag{4.9}$$

3. Multiplication by an Exponential Sequence

Multiplying $x[n]$ by $e^{j\omega_0 n}$ in time domain corresponds to a shift of ω_0 in the frequency domain:

$$e^{j\omega_0 n}x[n] \leftrightarrow X\left(e^{j(\omega-\omega_0)}\right) \tag{4.10}$$

4. Differentiation

Differentiating $X(e^{j\omega})$ with respect to ω corresponds to multiplying $x[n]$ by n :

$$nx[n] \leftrightarrow j\frac{dX(e^{j\omega})}{d\omega} \tag{4.11}$$

Proof :

$$-e^{j\omega}\frac{dX(e^{j\omega})}{de^{j\omega}} = -e^{j\omega}\frac{dX(e^{j\omega})}{d\omega}\cdot\left(\frac{de^{j\omega}}{d\omega}\right)^{-1} = j\frac{dX(e^{j\omega})}{d\omega} \quad (4.12)$$

5. Conjugation

The DTFT pair for $x^*[n]$ is given as:

$$x^*[n] \leftrightarrow X^*(e^{-j\omega}) \quad (4.13)$$

6. Time Reversal

The DTFT pair for $x[-n]$ is given as:

$$x[-n] \leftrightarrow X(e^{-j\omega}) \quad (4.14)$$

7. Convolution

If $(x_1[n], X_1(e^{j\omega}))$ and $(x_2[n], X_2(e^{j\omega}))$ are two DTFT pairs, then:

$$x_1[n] \otimes x_2[n] \leftrightarrow X_1(e^{j\omega})X_2(e^{j\omega}) \qquad (4.15)$$

In particular, for a LTI system with input $x[n]$, output $y[n]$ and impulse response $h[n]$, we have:

$$y[n] = x[n] \otimes h[n] \leftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}) \qquad (4.16)$$

which is analogous to (2.24) for continuous-time LTI systems

8. Multiplication

Multiplication in the time domain corresponds to convolution in the frequency domain:

$$x_1[n]\cdot x_2[n] \leftrightarrow X_1(e^{j\omega})\tilde{\otimes}X_2(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(e^{j\tau})X_2(e^{j(\omega-\tau)})d\tau \quad (4.17)$$

where $\tilde{\otimes}$ denotes convolution within one period

9. Parseval's Relation

The Parseval's relation addresses the energy of a sequence:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} \left|X(e^{j\omega})\right|^2 d\omega \quad (4.18)$$

With the use of (4.4), the proof is:

$$
\begin{aligned}
\sum_{n=-\infty}^{\infty} |x[n]|^2 &= \sum_{n=-\infty}^{\infty} x[n]x^*[n] \\
&= \sum_{n=-\infty}^{\infty} x[n] \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \right)^* \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) \left(\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right) d\omega \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| X(e^{j\omega}) \right|^2 d\omega
\end{aligned}
\tag{4.19}
$$

Part 2: Discrete Fourier Transform (DFT)

Definition

The discrete Fourier transform (DFT) may be regarded as a logical extension of the discrete-time Fourier transform (DTFT) covered earlier. By sampling the DTFT $X(e^{j\omega})$ at uniformly spaced frequencies $\omega_k = 2\pi k/N$, where $k = 0, 1, \ldots, N-1$, we can define the DFT of $x[n]$. Let $x[n]$, $n = 0, 1, \ldots, N-1$, be an N- point sequence. We define the DFT of $x[n]$ as

$$X[k] = \begin{cases} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} * & 0 \leq k \leq N-1 \\ 0, & \text{otherwise} \end{cases} \qquad (4.20)$$

The discrete Fourier transform relates discrete-time sequence $x[n]$, $n = 0, 1, \ldots, N - 1$, to the discrete-frequency sequence $X[k]$, $k = 0, 1, 2, \ldots, N - 1$. The discrete Fourier transform is defined by the pair DFT and the inverse discrete Fourier transform iDFT given as

$$x[n] = \begin{cases} \frac{1}{N} \sum_{k=0}^{N-1} X[k]\, e^{j2\pi kn/N} * & 0 \leq k \leq N - 1 \\ 0, & \text{otherwise} \end{cases} \tag{4.21}$$

Note that the DFT $X[k]$ as given in (7.1) is a periodic sequence with period N. This should be expected since $X[k]$ is a sampled version of the periodic DTFT $X(e^{j\omega})$. Also, note that both DFT and the inverse DFT (iDFT) begin the summation from 0 and end at $N - 1$ so that both are finite. Since DFT can be derived from DTFT, we would expect their properties to be similar. The properties of DFT are listed in the table.

Properties of the DFT

Property	Time Domain	Frequency Domain
1. Linearity	$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
2. Time-shifting	$x[n - m]$	$e^{-j2\pi km}X(k)$
3. Frequency-shifting (modulation)	$e^{-j\,2\pi k_0 n/N}x\,[n]$	$X(k - k_0)$
4. Time reversal	$x[-n]$	$X(-k)$
5. Conjugation	$x^*[n]$	$X^*(-k)$
6. Time-convolution	$x_1[n] \otimes x_2[n]$	$X_1[k]X_2[k]$
7. Frequency-convolution	$x_1[n]x_2[n]$	$\frac{1}{N}X_1[k] \otimes X_2[k]$
8. Parseval's relation	$E_x = \sum_{n=0}^{N-1} \lvert x[n] \rvert^2$	$E_x = \frac{1}{N}\sum_{k=0}^{N-1} \lvert X[k] \rvert^2$

Fast Fourier Transform (FFT)

The number of complex computations required in calculating DFT is drastically reduced by an algorithm known as the fast Fourier transform (FFT). The development of FFT, which dramatically reduces the number of computational operations, made DFT a very useful transform in many areas of applied science and technology. FFT was developed by James Cooley and John Tukey in 1965. Keep in mind that FFT is not a transform by itself; it is an algorithm—a fast computation of DFT.

It is evident from (4.20) that to compute each $X[k]$ requires N complex multiplications and $N - 1$ complex additions. Since we have to evaluate $X[k]$ for $k = 0, 1, 2, \ldots, N - 1$, the entire computation of DFT requires N^2 complex multiplications and $N(N - 1)$ complex additions. This becomes prohibitively time consuming for large N.

The numbers of complex multiplications required for N-point DFT (N^2) and FFT ($N/2 \log_2 N$) are compared below. We can see that as N increases, the speed advantage of FFT becomes obvious. Like inverse DFT (or iDFT), we also have inverse FFT (or iFFT). MATLAB/OCTAVE has in-built commands `fft` and `ifft` for finding respectively the fast Fourier transform and inverse fast Fourier transform of a discrete-time sequence.

Number of Multiplications Required in DFT and FFT

m	$N = 2^m$	DFT	FFT
1	2	4	1
2	4	16	4
3	8	64	12
4	16	256	32
5	32	1,024	80
6	64	4,096	192
7	128	16,384	448
8	256	65,536	1,024
9	512	261,144	2,304
10	1,024	1,048,576	5,120

Example 4.5

Find the DFT coefficients of a finite-duration sequence $x[n]$ which has the form of

$$x[n] = \begin{cases} 1, & n = 0, 1, 2 \\ 0, & \text{otherwise} \end{cases}$$

Using (4.20), we have:

$$X[k] = \sum_{n=0}^{2} x[n] W_N^{kn} = W_3^0 + W_3^k + W_3^{2k}$$
$$= e^{-\frac{j2\pi k}{3}} \left[1 + 2\cos\left(\frac{2\pi k}{3}\right) \right] = \begin{cases} 3, & k = 0 \\ 0, & k = 1, 2 \end{cases}$$

where $W_N = e^{-\frac{j2\pi}{N}}$

Together with $X[k]$ whose index is outside the interval of $0 \le k \le 2$, we finally have:

$$X[k] = \begin{cases} 3, & k = 0 \\ 0, & \text{otherwise} \end{cases}$$

If the length of $x[n]$ is considered as $N = 5$ such that $x[3] = x[4] = 0$, then we obtain:

$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn} = W_5^0 + W_5^k + W_5^{2k} \\ &= \begin{cases} e^{-\frac{j2\pi k}{5}} \left[1 + 2\cos\left(\frac{2\pi k}{5}\right)\right], & k = 0, 1, \cdots, 4 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

The MATLAB/OCTAVE command for DFT computation is `fft`. The code to produce magnitudes and phases of $X[k]$ is:

```
N=5;
x=[1 1 1 0 0];                    %append 2 zeros
subplot(2,1,1);
stem([0:N-1],abs(fft(x))); %plot magnitude response
title('Magnitude Response');
subplot(2,1,2);
stem([0:N-1],angle(fft(x)));%plot phase response
title('Phase Response');
```

The DFT will approach the DTFT when we append infinite zeros at the end of $x[n]$

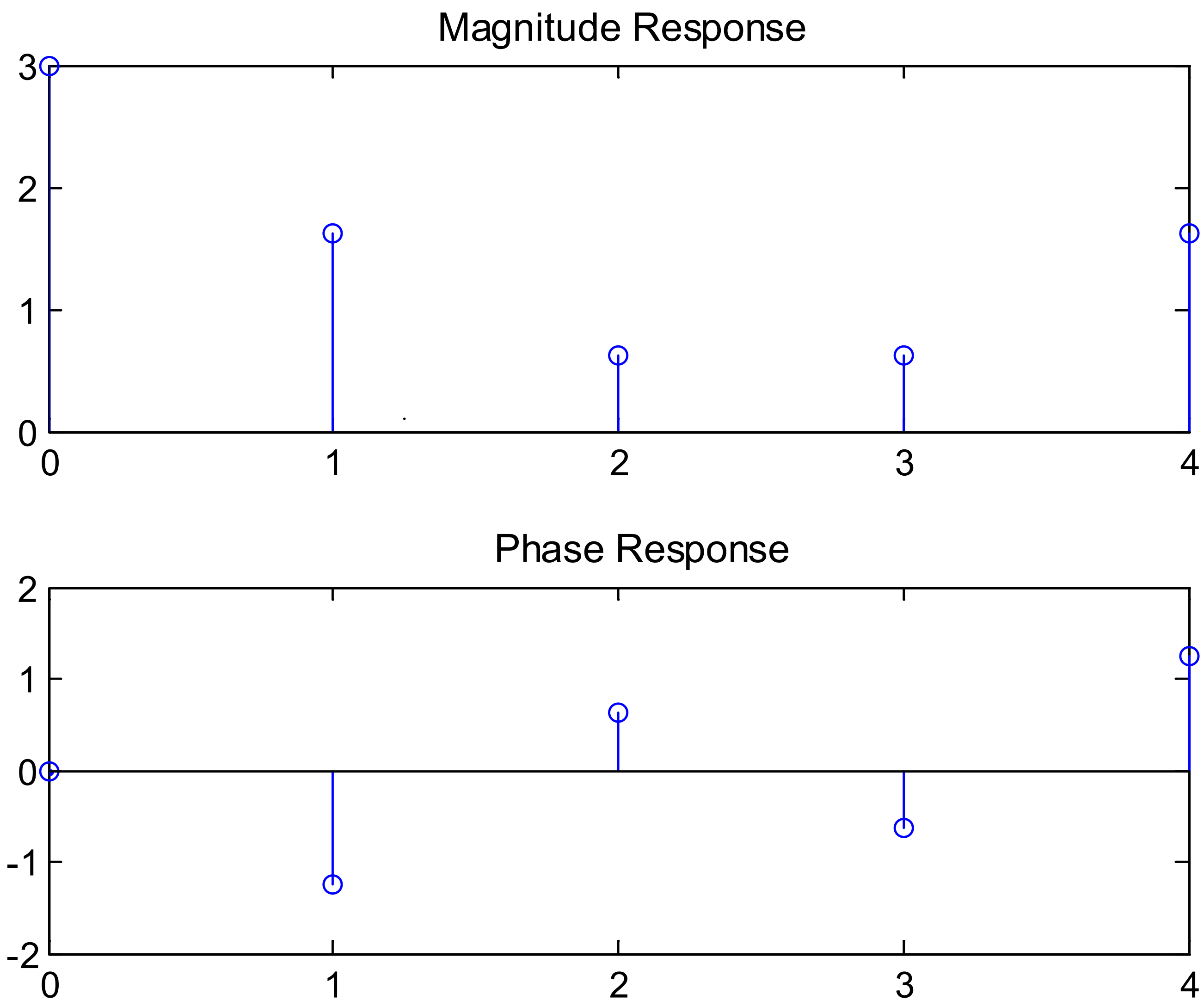

Fig. 4.3 DFT plots with $N = 5$

Example 4.6

Given a discrete-time finite-duration sinusoid:

$$x[n] = 2\cos(0.7\pi n + 1), \quad n = 0, 1, \cdots, 20$$

Estimate the tone frequency using DFT.

Consider the continuous-time case first. According to (2.16), Fourier transform pair for a complex tone of frequency Ω_0 is:

$$e^{j\Omega_0 t} \leftrightarrow 2\pi\delta(\Omega - \Omega_0)$$

That is, Ω_0 can be found by locating the peak of the Fourier transform. Moreover, a real-valued tone $\cos(\Omega_0 t)$ is:

$$\cos(\Omega_0 t) = \frac{e^{j\Omega_0 t} + e^{-j\Omega_0 t}}{2}$$

From the Fourier transform of $\cos(\Omega_0 t)$, Ω_0 and $-\Omega_0$ are located from the two impulses.

Analogously, there will be two peaks which correspond to frequencies 0.7π and -0.7π in the DFT for $x[n]$.

The MATLAB code is

```
N=21;                  %number of samples is 21
A=2;                   %tone amplitude is 2
w=0.7*pi;              %frequency is 0.7*pi
p=1;                   %phase is 1
n=0:N-1;               %define a vector of size N
x=A*cos(w*n+p);        %generate tone
X=fft(x);              %compute DFT
subplot(2,1,1);
stem(n,abs(X));       %plot magnitude response
subplot(2,1,2);
stem(n,angle(X));     %plot phase response
```

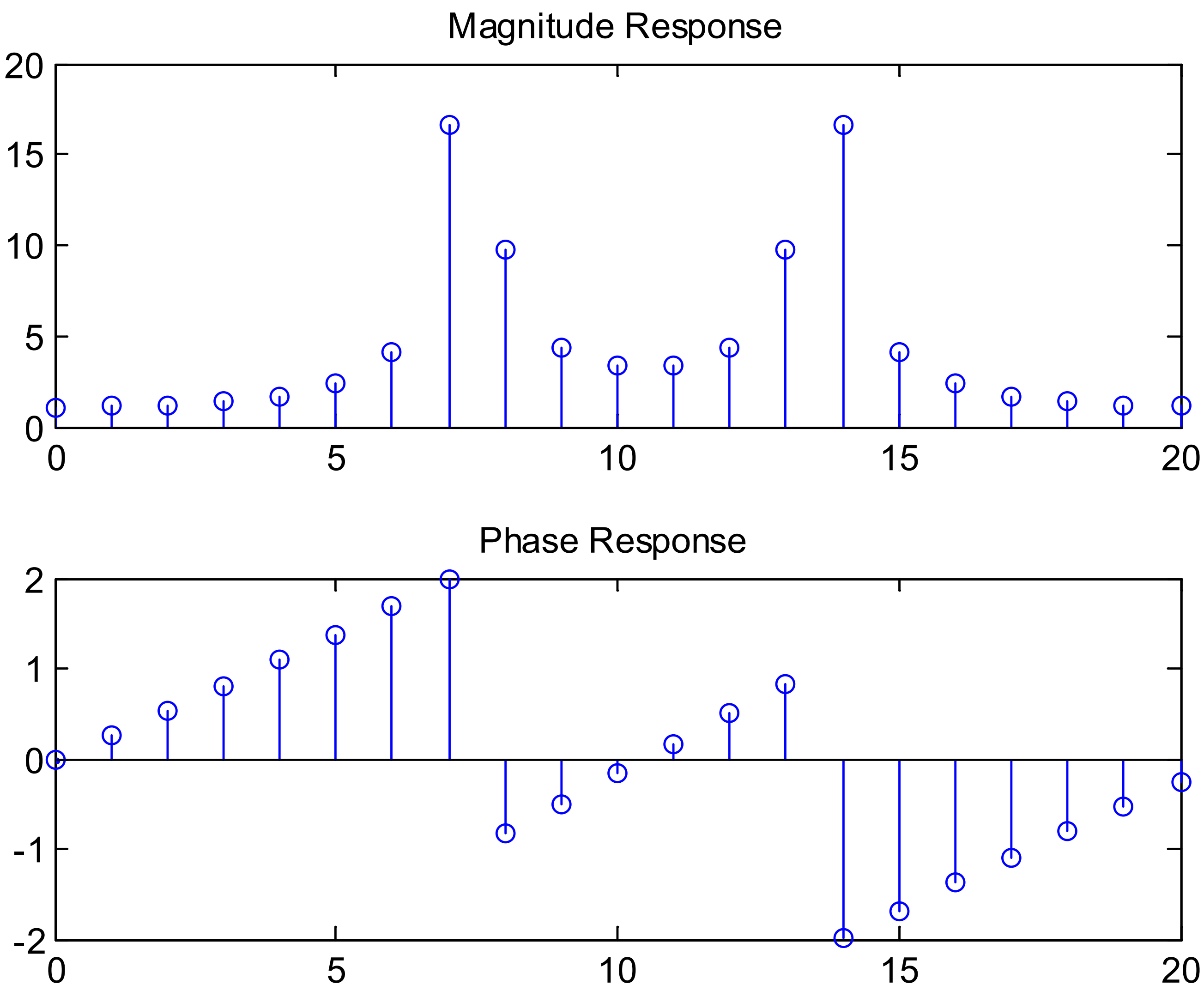

Fig. 4.4 DFT plots for a real tone

```
X =
 1.0806              1.0674+0.2939i   1.0243+0.6130i
 0.9382+0.9931i   0.7756+1.5027i   0.4409+2.3159i
-0.4524+4.1068i  -6.7461+15.1792i  6.5451-7.2043i
 3.8608-2.1316i   3.3521-0.5718i   3.3521+0.5718i
 3.8608+2.1316i   6.5451+7.2043i  -6.7461-15.1792i
-0.4524-4.1068i   0.4409-2.3159i   0.7756-1.5027i
 0.9382-0.9931i   1.0243-0.6130i   1.0674-0.2939i
```

Interestingly, we observe that $\Re\{X[k]\} = \Re\{X[N-k]\}$ and $\Im\{X[k]\} = -\Im\{X[N-k]\}$. In fact, all real-valued sequences possess these properties so that we only have to compute around half of the DFT coefficients.

As the DFT coefficients are complex-valued, we search the frequency according to the magnitude plot.

There are two peaks, one at $k=7$ and the other at $k=14$ which correspond to $\omega=0.7\pi$ and $\omega=-0.7\pi$, respectively.

From Example 4.2, it is clear that the index k refers to $\omega=2\pi k/N$. As a result, an estimate of ω_0 is:

$$\hat{\omega}_0 = \frac{2\pi \cdot 7}{21} \approx 0.6667\pi$$

To improve the accuracy, we append a large number of zeros to $x[n]$. The MATLAB code for $x[n]$ is now modified as:

```
x=[A*cos(w.*n+p) zeros(1,1980)];
```

where 1980 zeros are appended.

The MATLAB code is provided as `ex6_6.m` and `ex6_6_2.m`.

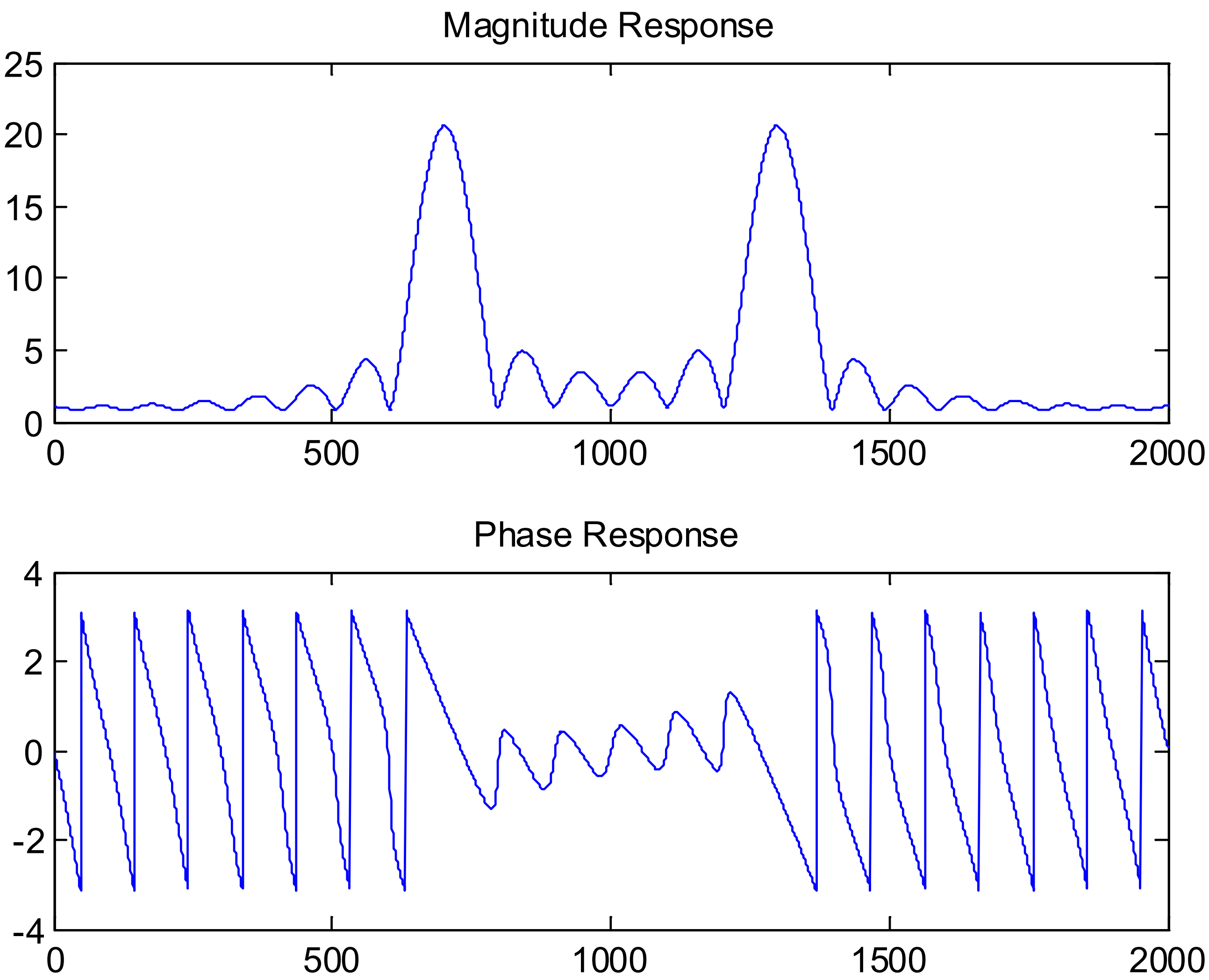

Fig. 4.5 DFT plots for a real tone with zero padding

The peak index is found to be $k = 702$ with $N = 2001$. Thus

$$\hat{\omega}_0 = \frac{2\pi \cdot 702}{2001} \approx 0.7016\pi$$

<u>Example 4.7</u>

Find the inverse DFT coefficients for $X[k]$ which has a length of $N = 5$ and has the form of

$$X[k] = \begin{cases} 1, \ n = 0, 1, 2 \\ 0, \ n = 3, 4 \end{cases}$$

Plot $x[n]$.

Using (4.21) and Example 4.5, we have:

$$x[n] = \frac{1}{N}\sum_{n=0}^{N-1} X[k]W_N^{-kn} = \frac{1}{5}\left(W_5^0 + W_5^{-n} + W_5^{-2n}\right)$$

$$= \begin{cases} \frac{1}{5}e^{\frac{j2\pi n}{5}}\left[1 + 2\cos\left(\frac{2\pi n}{5}\right)\right], & n = 0, 1, \cdots, 4 \\ 0, & \text{otherwise} \end{cases}$$

The main MATLAB code is:

```
N=5;
X=[1 1 1 0 0];
subplot(2,1,1);
stem([0:N-1],abs(ifft(X)));
subplot(2,1,2);
stem([0:N-1],angle(ifft(X)));
```

The MATLAB program is provided as `ex6_7.m`.

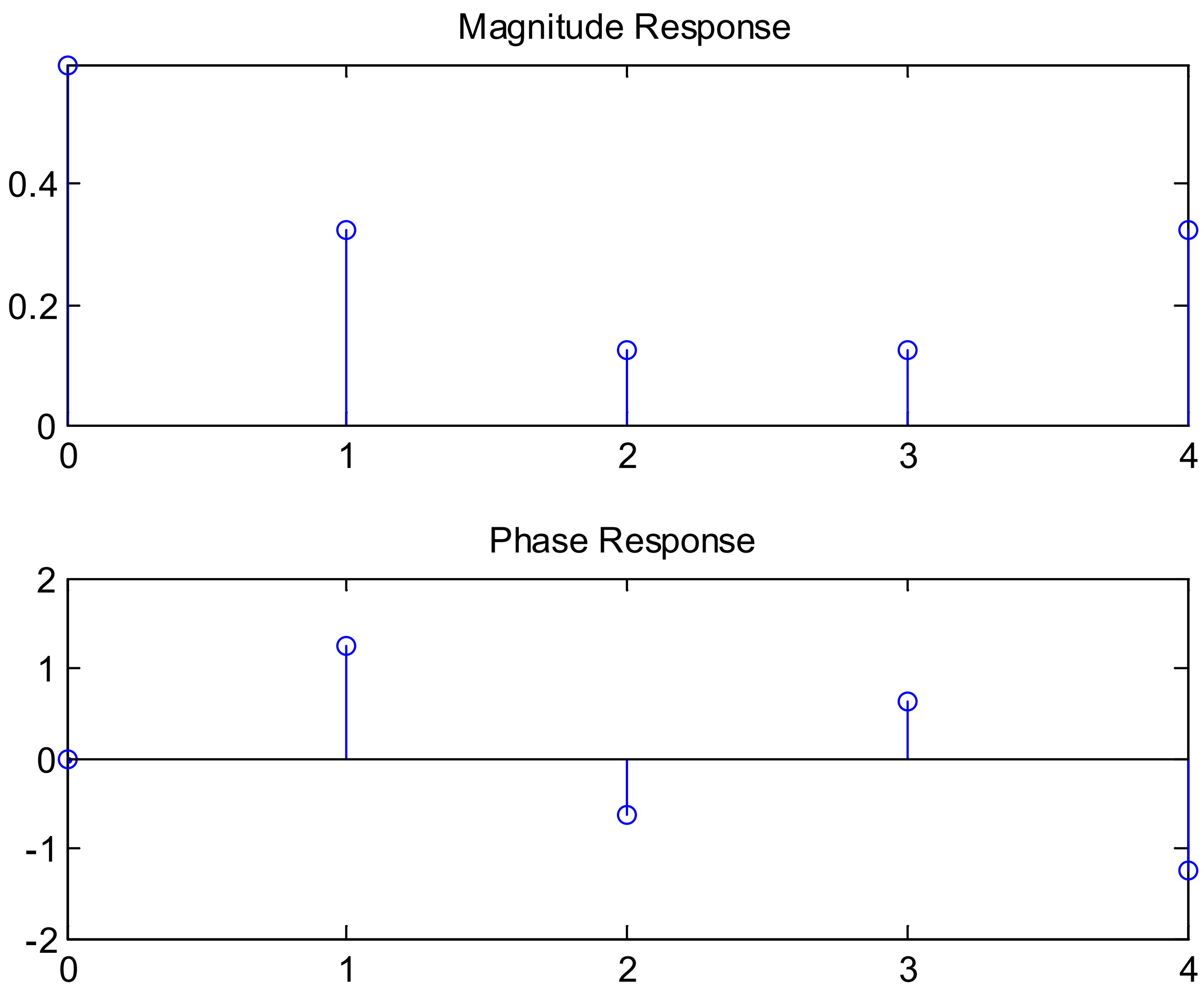

Fig. 4.6 Inverse DFT plots